BioHackathon series:
BioHackathon SWAT4HCLS 2026
Amsterdam, The Netherlands, 2026
Project 8

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Introduction

As part of the SWAT4HCLS 2026 BioHackathon, we here report our work on the design and implementation of a tool that serves a virtual flat-file view over data stored in a relational representation and demonstrated using Variant Call Format (VCF) genomic data files.

Meeting information

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biohackathon_location: "Amsterdam, The Netherlands, 2026"
group: Project 8
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```

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BioHackathons is about hacking together, and the minimal number of authors for reports is two. This makes a minimal example look like this:

```
authors:
  - name: Elias Crum
    orcid: 0009-0005-3991-754X
    affiliation: [1,2]
    role: VCF to TSV conversion, duckDB data ingestion, manuscript writing + editi
  - name: Benno Kruit
    orcid: 0000-0000-0000-0000
    affiliation: 3
    role: FUSE virtual file generation from duckDB data, manuscript writing + edit
affiliations:
  - name: IDLab, Department of Electronics and Information Systems, Ghent Universi
    index: 1
```

```yaml
  - name: Flemish institute for Technological Research (VITO) Mol, Belgium
    index: 2
  - name: example
    index: 3
```


Introduction

Introduction

This report describes the design and implementation of a tool that serves a virtual flat-file view over data stored in a relational representation. The central idea is to preserve the queryability and compact storage of a tabular backend while exposing the data through a conventional file interface that can be inspected with standard filesystem operations. In this work, compressed storage is provided through Apache Parquet, a columnar format designed for efficient storage and retrieval with built-in compression and encoding support, while DuckDB is used as the embedded analytical database managing the relational view of the data. The file-oriented presentation layer is implemented with FUSE, which enables user-space programs to export filesystems to the kernel and thereby present generated content as ordinary files. :contentReferenceoaicite:0

As a concrete demonstration, the tool is applied to genomic variant data represented in Variant Call Format (VCF). VCF is a text-based format consisting of meta-information lines, a header, and tab-delimited variant records with genotype information for one or more samples, making it a natural target for virtual flat-file materialization from relational tables. This setting provides a useful test case because VCF combines structured schema-like fields with a strict textual serialization, allowing the system to illustrate how relational records can be reconstructed into a standards-compliant, human-readable file without duplicating the underlying dataset. :contentReferenceoaicite:1

Formatting

This document use Markdown and you can look at this tutorial.

Subsection level 2

Please keep sections to a maximum of only two levels.

Tables

Tables can be added in the following way, though alternatives are possible:

Table: Note that table caption is automatically numbered and should be given before the table itself.

```
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| -------- | -------- |
| item 1 | item 2 |
| item 3 | item 4 |
```

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item 1	item 2
item 3	item 4

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Other main section on your manuscript level 1

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1. Item 1
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Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- citesAsDataSource: when you point the reader to a source of data which may explain a claim

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- providesDataFor
- obtainsSupportFrom
- discusses
- extends
- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

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Acknowledgements

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References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, *32*(14), 2236–2238. https://doi.org/10.1093/bioinformatics/btw214 **[cito:citesAsAuthority]**

Author contributions

Elias Crum: VCF to TSV conversion, duckDB data ingestion, manuscript writing + editing
Benno Kruit: FUSE virtual file generation from duckDB data, manuscript writing + editing